

M42 — Inline device manager in the encounter console

Phase: Phase 7 follow-on (post-M41, operator-feedback fix) **Status:** **DONE** (Phase A — inline embed; Phase B/C deferred to M43) **Blocked by:** M2 (Encounter dataclass + `get_by_join_code`), M22 (Per-Patient Console), M39 (modal-dialog pattern reused here) **Blocks:** M43 (shared med carts + grouped MAR — explicitly deferred) **Estimated effort:** 0.5 day

1. Purpose

Operator feedback after M41:

"On each encounter the device manager should be setup to operate within the encounter like it does for the single patient not having to leave the encounter through a link like it currently is. All the device functions and assignments should function the same. The assignment for a device being added should pre-populate with the patient Character of the encounter."

Three problems wrapped in one report:

1. **The link-out was broken in multi-patient mode.** The encounter console's Devices card linked to `/portal/control/ops?join=...`, but that route used `control_session.get_active()` — which per the M2 contract returns `None` in any multi-encounter room. The instructor clicked the link and got bounced to the wizard. 2. **The link itself was the wrong UX.** Even when it worked (single-patient mode), it pulled the operator out of the encounter console — losing telemetry, ECG, scene injector, transcript pane context. 3. **The add-device modal hard-coded the patient assignment to "— unassigned —"** (see `control_ops_devices.js:272: fillCharacterSelect($('ad-char'), '')`). Even in single-patient mode the operator had to re-pick the patient every time.

M42 fixes all three with three small changes:

1. `/portal/control/ops` now accepts `?join=<code>` and resolves the session via `control_session.get_by_join_code` first, falling back to `get_active()` for v6-compat. Works in multi-patient mode. 2. The ops route also takes `?patient_persona_id=<pid>` (override) and `?embed=1` (hides the ops-view header for iframe embedding). Falls back to the session's primary patient when no override is passed. 3. The encounter console's Devices card replaces the link-out with a **Manage devices** button that opens a modal dialog containing the ops view in an iframe — same pattern as M39's Engage modal. The iframe URL carries `?join=&embed=1` so the ops view renders scoped + chromeless; popout link keeps the non-embed URL for second-monitor workflows.

Explicitly out of scope (deferred to M43):

- **Shared med carts across multiple encounters.** Today a cabinet

(`device_kind="cabinet", device_model="pyxis"`) has a single `character_id` field — one patient per cart. The user wants one cart shared across multiple beds.

- **Medicine administration grouped by patient character.** Today

the cart's MAR is seeded once per session; v7's per-patient grouping is a model change that depends on the shared-cart work above.

These two pieces are bigger architectural changes (model migration + MAR rewrite) and would have ballooned M42 past one module. M43 will pick them up.

2. Structure

Files touched:

- `portal/server.py` — `/portal/control/ops` route:
- New query params: `join`, `patient_persona_id`, `embed`.
- Resolution: `control_session.get_by_join_code(join) → fallback`

to `control_session.get_active()` → fallback redirect. - Renders the template with two new keys: `default_device_patient_id` (resolved from query param OR `sess.patient_persona_id` OR the session's first selected persona) and `embed_mode` (bool).

- `portal/templates/control_ops.html`:

- Bootstrap `window.MEDSIM2_OPS` now carries

`default_device_patient_id` and `embed_mode`. - When `embed_mode` is true, a `<style>` block hides the v6 ops-view chrome (`.topbar`, `.ops-header`, `.control-bar`, `.session-context-card`, `.live-transcript-card`, `.operator-ptt-card`, `.seed-report-card`, `.medication-checklist-card`). The device manager, invite- stations card, and connected-stations card stay visible.

- `portal/static/control_ops_devices.js` — `openAddDevice()` now

reads `window.MEDSIM2_OPS.default_device_patient_id` and passes it to `fillCharacterSelect($('ad-char'), defaultPatient)` instead of the hard-coded empty string. Falls back to `' '` when no default is set (v6 single-patient behavior unchanged).

- `portal/templates/encounter_console.html`:

- Devices card's link-out swapped for a `<button id="btn-manage-`

`devices">` styled as a primary blue pill. - New `<dialog id="devices-dialog">` near the engage-dialog with `iframe` + close button + popout anchor. Iframe starts at `about:blank`.

- `portal/static/encounter_console.js` — new `DOMContentLoaded`

handler binds `#btn-manage-devices` to opening the modal. Sets the `iframe src` to `/portal/control/ops?join=<jc>&embed=1` and the popout anchor's `href` to the same URL without `embed=1`. Close button + native close event blank the `iframe` (same cleanup pattern as M39).

- `portal/static/encounter_console.css` — `.device-manage-btn`

styled as a primary action.

No backend dataclass change. No schema migration. The `DeviceStation` model is untouched in M42 (M43 will change it for shared carts).

3. Uses

3.1 Operator flow

1. Operator is on a Per-Patient Console (`/portal/room/encounter/{id}`). 2. Scrolls to the Devices card. Sees the bound-device list (M22 summary) + a **Manage devices** button. 3. Clicks the button. Modal dialog opens overlaying the encounter console. Backdrop dims the underlying console. 4. Modal `iframe` loads `/portal/control/ops?join=<encounter join>& embed=1`. The ops view's top header + transcript + ptt panel are hidden (embed mode). The instructor sees: invite-stations card (chat + EHR QR), connected-stations roster, and the device manager — exactly the surfaces relevant to device management. 5.

Operator clicks **+ Add device** inside the iframe. The patient dropdown is **pre-selected to this encounter's primary patient** (e.g. *Mr. Hayes (P-014)*) — no need to re-pick. 6. Operator picks kind/model, optionally tweaks the patient assignment, hits Mint QR. Device registers via the existing `POST /api/device/register`. Modal stays open; new device appears in the iframe's roster. 7. Operator closes the modal (**X Close** button or `ESC`). Iframe src blanks (cleans up any in-flight WS connections); operator lands back on the encounter console. 8. The bound device shows up in the encounter console's Devices card on the next 2s `/api/room/state` poll.

3.2 Why iframe-embed vs. native inline

Native inline would require porting ~1500 LOC of working v6 device-management JS into the encounter console namespace. That's a multi-day refactor with real risk of behavior drift. The iframe-embed approach reuses every line of that code unchanged — the only changes are the route's lookup path (one query param) and the add-device default (one constant). Same trade-off M39 made for the engage chat UI.

3.3 Behavior matrix

Path	Pre-M42	Post-M42
Single-patient ops view (<code>/portal/control/ops</code>)	works (<code>get_active</code>)	works (<code>get_active</code> fallback)
Multi-patient ops link-out from encounter console	broken (<code>get_active=None</code> → 303 to wizard)	works (<code>?join=...</code> lookup)
Add-device patient dropdown default	hard-coded "" (unassigned)	encounter's primary persona
Operator stays in encounter window	no — link-out new tab	yes — modal overlay

4. Functions (exported API surface)

Symbol	Where	Purpose
<code>GET /portal/control/ops</code>	<code>portal/server.py</code>	Now accepts <code>?join=<code> + ?patient_persona_id=<pid> + ?embed=1</code> . Backward-compatible with the v6 callsites.
<code>window.MEDSIM2_OPS.default_device_patient_id</code>	<code>portal/templates/control_ops.html</code>	Bootstrap field. Empty string when unscoped.
<code>window.MEDSIM2_OPS.embed_mode</code>	same	Bootstrap field. True when iframed.
<code>openAddDevice()</code> (updated)	<code>portal/static/control_ops_devices.js</code>	Reads <code>default_device_patient_id</code> from bootstrap; default arg of <code>fillCharacterSelect</code> is no longer hard-coded "".
Modal markup	<code>portal/templates/encounter_console.html</code>	<code><dialog id="devices-dialog"> + iframe</code> .
Button wiring	<code>portal/static/encounter_console.js</code>	<code>DOMContentLoaded</code> handler binds <code>#btn-manage-devices</code> → opens modal with the scoped iframe URL.

5. Limitations

- **Cabinets are still single-patient.** Med carts (``device_model=`

`"pyxis")` carry one `character_id`. Shared-cart support requires changing the `DeviceStation`` model + the cabinet's bootstrap + the cart's MAR rendering. M43 will pick this up.

- **The grouped per-patient MAR view doesn't exist yet.** Today

the cabinet UI lists meds as a flat per-station list. Operators asked for `"each of the assigned character patients and their assigned medication under their character name"` — also M43.

- **Embed-mode hides the v6 PTT panel + transcript.** Those have

v7-native equivalents on the encounter console already (the voice card with Engage modal from M33/M39, and the live transcript card). Hiding them in the iframe is a feature, not a regression.

- **The CSS-based header hide could miss future v6 ops-view**

elements. If a future v6 change adds a new top-of-page element, it'll show up in embed mode until we list it in the hide block. Acceptable for now; easy to extend.

- **The patient pre-populate uses the encounter's**

`patient_persona_id`. If the operator changes the primary patient on the encounter (no UI for that today), already-bound devices keep their old `character_id`. New devices get the new default. Matches the snapshot-on-create v6 behavior.

- **The modal's iframe sandbox:** we don't set a `sandbox`

attribute. The iframe runs same-origin so cookies + WebSockets work for the device manager. If a future security review wants isolation, the iframe's behavior would need to switch to a cross-origin allowlist.

- **No keyboard shortcut to open the device manager.** Click

only. Out of scope for v7.0.

- **The button label is "Manage devices"** (not "Manage devices

and medications") even though the modal also exposes the medication checklist. Acceptable — meds-as-devices is a fuzzy line.

6. Test status

Test file	Asserts	Status	Last run
<code>tests/v7/test_inline_device_manager.py::test_ops_view_loads_via_join_code_in_multi_encounter_room</code>	<code>/portal/control/ops?join=<bed join></code> returns 200 in a 2-encounter room (broken pre-M42)	PASS	2026-05-27
<code>...::test_ops_view_with_unknown_join_falls_back_to_wizard</code>	Unknown join → 303 to <code>/portal/control</code> (no <code>get_active()</code> session)	PASS	2026-05-27
<code>...::test_ops_view_bootstrap_pre_fills_patient_persona_from_query</code>	<code>?patient_persona_id=P-014</code> → bootstrap carries <code>default_device_patient_id=P-014</code>	PASS	2026-05-27

Test file	Asserts	Status	Last run
...::test_ops_view_bootstrap_falls_back_to_session_primary	Without override, defaults to session's patient persona	PASS	2026-05-27
...::test_ops_view_embed_mode_hides_top_header	?embed=1 injects a <style> block hiding .topbar etc	PASS	2026-05-27
...::test_device_js_uses_default_patient_on_add_device	openAddDevice() reads default_device_patient_id and passes it to fillCharacterSelect	PASS	2026-05-27
...::test_encounter_console_renders_manage_devices_button	Devices card has #btn-manage-devices; old link-out is gone	PASS	2026-05-27
...::test_encounter_console_devices_modal_present	<dialog id="devices-dialog"> + iframe + close + popout markup present; iframe starts at about:blank	PASS	2026-05-27
...::test_devices_dialog_popout_href_is_scoped_to_encounter	Popout anchor href has join=<code> + patient_persona_id=<pid>	PASS	2026-05-27
...::test_encounter_console_js_wires_manage_devices_button	encounter_console.js binds the button, builds the embed URL, blanks the frame on close	PASS	2026-05-27
Full v7 suite	284 passed, 1 skipped (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-codre	Initial M42 (Phase A): ops route accepts ?join=?patient_persona_id+?embed; bootstrap carries them; device-add JS reads the patient default; encounter console gets a #btn-manage-devices button + <dialog id="devices-dialog"> + iframe modal; CSS for the button; 10 new tests	portal/server.py, portal/templates/control_ops.html, portal/static/control_ops_devices.js, portal/templates/encounter_console.html, portal/static/encounter_console.{js,css}, tests/v7/test_inline_device_manager.py (new)

8. Open questions / known issues

- **M43 — Shared med carts across encounters.** The DeviceStation

model needs a character_ids: list[str] field (or a separate cabinet_assignments table linking cart_id ↔ encounter_id). The cart's bootstrap response (/api/device/<sid>/bootstrap) needs to return a list of patient MARs, not one. The cabinet UI needs a per-patient section header.

- **M43 — Grouped MAR view.** Each patient gets a labeled section

in the cart UI showing their meds (name, dose, route, status). Dispensing logs to the per-patient `chart_event` with the current patient context. Migration: existing single-patient carts become single-element lists transparently.

- **M43 — UI for shared-cart assignment.** A new "+ Assign

additional patient" button on the device-detail modal would add another encounter to the cart's `character_ids`. Today's M42 inline manager will show the M43 capability automatically once the backend ships.

- **Embed-mode CSS coverage.** The hide list in

`control_ops.html`'s `{% if embed_mode %}` block hides what's visible today. A future operator-facing review (LAN test) may surface other v6 elements that should be hidden in the modal.

- ****The popout anchor is generated server-side with the encounter's**

`patient_persona_id` baked in.** If the operator later changes the primary patient (no UI today), the popout link stales until page refresh. Acceptable; documented.

- **Browser back-button inside the iframe.** If the operator

navigates within the iframe and then hits browser back, behavior depends on the browser's iframe history isolation. Verify in LAN test if anyone hits this.

Render this guide to PDF with `python docs/module_guides/render_pdfs.py`.